

SECTION 5: Electric Dipole & Dipole Moment (Q51–60)

51. The SI unit of dipole moment is:

Answer: A) C·m

Explanation: Dipole moment $\mathbf{p} = q \times d$ has units **Coulomb × meter**.

52. The torque on a dipole in uniform electric field is maximum when:

Answer: B) $\theta = 90^\circ$

Explanation: Torque $\tau = pE \sin \theta$ is maximum at $\theta = 90^\circ$.

53. Dipole moment direction is from:

Answer: B) Negative to positive

Explanation: By convention, the dipole moment vector points **from negative to positive charge**.

54. Electric field along the axial line of a dipole is:

Answer: B) Twice the equatorial field

Explanation: On axial line $E_{\text{axial}} = \frac{2p}{4\pi\epsilon_0 r^3}$; equatorial $E_{\text{equatorial}} = \frac{p}{4\pi\epsilon_0 r^3}$.

55. Which of the following is an example of permanent dipole?

Answer: A) HCl molecule

Explanation: HCl has a **permanent separation of charge** due to difference in electronegativity.

56. Torque on a dipole is given by:

Answer: B) $p \cdot E \cdot \sin \theta$

Explanation: Torque formula: $\tau = pE \sin \theta$

57. Dipole in non-uniform field experiences:

Answer: C) Both force and torque

Explanation: Non-uniform field → net **force** and **torque**; uniform field → only torque.

58. For an ideal dipole, charge separation is:

Answer: C) Very small

Explanation: Ideal dipole: charges are equal and opposite with **infinitesimally small separation**.

59. Electric field on equatorial line is:

Answer: B) Perpendicular to axis

Explanation: Field direction along equatorial line is **perpendicular** to dipole axis.

60. Dipole in uniform field has potential energy:

Answer: A) $U = -\mathbf{p} \cdot \mathbf{E} \cdot \cos \theta$

Explanation: $U = -\mathbf{p} \cdot \mathbf{E} = -pE \cos \theta$

SECTION 6: Electric Potential & Potential Difference (Q61–75)

61. Electric potential is:

Answer: B) Work done per unit charge

Explanation: $V = W/q$

62. SI unit of electric potential:

Answer: A) Volt

Explanation: Volt = 1 Joule/Coulomb

63. Potential difference between two points A and B is:

Answer: A) W/q

Explanation: $V = \frac{W}{q}$

64. Work done in moving a charge along equipotential surface:

Answer: B) Zero

Explanation: No work is done along **equipotential**, as potential difference = 0.

65. Potential at infinity due to a point charge is:

Answer: A) Zero

Explanation: By convention, $V(\infty) = 0$

66. Potential is a:

Answer: B) Scalar

Explanation: Potential has magnitude only, no direction.

67. Electric potential at distance r from point charge q:

Answer: B) $V = kq/r$

Explanation: $V = k \frac{q}{r}$

68. If potential difference is doubled, energy of a charge:

Answer: A) Doubles

Explanation: $U = qV$, doubling $V \rightarrow U$ doubles.

69. Relation between electric field E and potential V:

Answer: A) $E = -dV/dr$

Explanation: Electric field is **negative gradient of potential**.

70. Potential due to dipole at equatorial line:

Answer: A) Zero

Explanation: Contributions from +q and -q cancel along equatorial line.

71. Moving a charge along equipotential surface requires:

Answer: B) Zero work

Explanation: No potential difference $\rightarrow W = 0$.

72. Electric potential inside a conductor:

Answer: C) Constant

Explanation: Electric field inside conductor = 0 $\rightarrow$ potential is **constant**.

73. Relation between electric field and potential difference in uniform field:

Answer: A) $E = V/d$

Explanation: For uniform field $E = \frac{V}{d}$

74. Unit of potential difference is:

Answer: B) Volt

Explanation: Standard SI unit.

75. Potential energy of a charge q at potential V:

Answer: A) $U = qV$

Explanation: By definition of electric potential energy.

SECTION 7: Capacitance & Combination of Capacitors (Q76–95)

76. Capacitance is defined as:

Answer: A) Q/V

Explanation: $C = \frac{Q}{V}$

77. SI unit of capacitance:

Answer: A) Farad

Explanation: $1 \text{ F} = 1 \text{ C/V}$

78. Capacitance of parallel plate capacitor depends on:

Answer: A) Area of plates & separation

Explanation: $C = \epsilon_0 A/d$

79. For capacitors in series, equivalent capacitance is:

Answer: B) Reciprocal of sum of reciprocals

Explanation: $1/C_{\text{eq}} = 1/C_1 + 1/C_2$

80. For capacitors in parallel, equivalent capacitance is:

Answer: A) Sum of capacitances

Explanation: $C_{\text{eq}} = C_1 + C_2$

81. Energy stored in a capacitor:

Answer: A) $\frac{1}{2} CV^2$

Explanation: Standard formula: $U = \frac{1}{2} CV^2$

82. Dielectric increases capacitance because:

Answer: D) All of above

Explanation: Reduces potential difference, reduces field, increases charge storage.

83. Dielectric constant of vacuum:

Answer: A) 1

Explanation: Vacuum is reference; $K = 1$

84. Capacitance of spherical capacitor depends on:

Answer: A) Radii of shells

Explanation: $C = 4\pi\epsilon_0 \frac{ab}{b-a}$

85. Capacitance of parallel plate capacitor is directly proportional to:

Answer: B) Area of plates

Explanation: $C \propto A$

86. Capacitance of parallel plate capacitor is inversely proportional to:

Answer: A) Plate separation

Explanation: $C \propto 1/d$

87. Combination of two capacitors, 4 μF and 6 μF in series:

Answer: B) 2.4 μF

Explanation: $1/C = 1/4 + 1/6 = 5/12 \rightarrow C = 12/5 = 2.4 \mu\text{F}$

88. Combination of two capacitors, $4 \mu\text{F}$ and $6 \mu\text{F}$ in parallel:

Answer: A) $10 \mu\text{F}$

Explanation: $C_{\text{eq}} = 4 + 6 = 10 \mu\text{F}$

89. Energy stored in $2 \mu\text{F}$ capacitor at 10 V :

Answer: B) 0.5 mJ

Explanation: $U = \frac{1}{2}CV^2 = 0.5 * 2e-6 * 100 = 1e-4 \text{ J} = 0.5 \text{ mJ}$

90. Adding dielectric in capacitor:

Answer: B) Reduces energy at constant voltage

Explanation: $U = \frac{1}{2}CV^2$, C increases $\rightarrow U$ decreases if V constant.

91. Parallel plate capacitor has area 0.02 m^2 , separation 1 mm , $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$:

Answer: C) $1.77 \times 10^{-10} \text{ F}$

Explanation: $C = \epsilon_0 A/d = 8.85e-12 * 0.02/1e-3 \approx 1.77e-10 \text{ F}$

92. Capacitance depends on:

Answer: B) Geometry & dielectric

Explanation: Depends on A , d , and dielectric.

93. Two identical capacitors connected in series: total capacitance is:

Answer: A) Half of one

Explanation: $C_{\text{series}} = C/2$

94. Total charge on series combination is:

Answer: A) Same on each

Explanation: Series $\rightarrow$ same charge on each capacitor.

95. Capacitance can be increased by:

Answer: B) Using dielectric

Explanation: Dielectric increases C by factor K .

SECTION 8: Gauss's Law & Electric Flux (Q96–120)

96. Gauss's law relates:

Answer: B) Electric flux & enclosed charge

Explanation: $\Phi_E = Q_{\text{enclosed}}/\epsilon_0$

97. Flux through closed surface enclosing charge Q :

Answer: A) Q/ϵ_0

Explanation: Direct from Gauss's law.

98. Electric field inside conductor in electrostatic equilibrium:

Answer: A) Zero

Explanation: Free charges redistribute $\rightarrow E = 0$.

99. Electric field due to infinite plane sheet:

Answer: B) Constant

Explanation: $E = \sigma/2\epsilon_0$ independent of distance.

100. Electric field due to infinite line charge:

Answer: B) $1/r$

Explanation: Cylindrical symmetry $\rightarrow E = \lambda/2\pi\epsilon_0 r$

101. Gaussian surface should be:

Answer: B) Closed

Explanation: Gauss's law applies only for closed surfaces.

102. Flux through closed surface with no enclosed charge:

Answer: A) Zero

Explanation: $\Phi = Q_{\text{enclosed}}/\epsilon_0 = 0$

103. Flux is:

Answer: A) Scalar

Explanation: Flux = $E \cdot A$ scalar.

104. Electric field outside uniformly charged sphere:

Answer: A) Like point charge

Explanation: By symmetry, $E = kQ/r^2$ outside sphere.

105. Electric flux depends on:

Answer: D) All above

Explanation: Magnitude, area, and angle matter $\rightarrow \Phi = E \cdot A \cos \theta$

106. Unit of electric flux:

Answer: A) $\text{N}\cdot\text{m}^2/\text{C}$

107. Which surface is best for Gauss's law calculation?

Answer: B) Symmetric

Explanation: Symmetry simplifies calculations.

108. Flux through cube enclosing charge Q:

Answer: A) Q/ϵ_0

Explanation: Orientation doesn't matter for closed surface.

109. Field inside hollow conductor:

Answer: A) Zero

110. Field inside solid conductor:

Answer: A) Zero

111. Gauss's law is a consequence of:

Answer: A) Coulomb's law

112. Electric flux through plane perpendicular to field:

Answer: A) EA

113. Flux through plane parallel to field:

Answer: B) Zero

114. Electric field due to thin charged shell inside:

Answer: A) Zero

115. Electric field due to thin charged shell outside:

Answer: A) Like point charge

116. For non-conducting solid sphere, field at center:

Answer: A) Zero

117. Field inside spherical shell:

Answer: A) Zero

118. Field at surface of charged sphere:

Answer: D) Both A & C

119. Electric flux is positive if field:

Answer: A) Exits surface

120. Flux through cube with no enclosed charge:

Answer: A) Zero